Highest common factor

1 Consider the following prime factorisations:

$$180 = 2 \times 2 \times 3 \times 3 \times 5$$

$$600 = 2 \times 2 \times 2 \times 3 \times 5 \times 5$$

Find the highest common factor of 180 and 600.

2 Consider the following prime factorisations:

$$224 = 2 \times 2 \times 2 \times 2 \times 2 \times 7$$

$$196 = 2 \times 2 \times 7 \times 7$$

Find the highest common factor of 224 and 196.

3 Consider the following prime factorisations:

$$4900 = 2 \times 2 \times 5 \times 5 \times 7 \times 7$$

$$1750 = 2 \times 5 \times 5 \times 5 \times 7$$

Find the highest common factor of 4900 and 1750.

4 Consider the following prime factorisations:

$$140 = 2^2 \times 5 \times 7$$

$$24\,500 = 2^2 \times 5^3 \times 7^2$$

Find the highest common factor of 140 and $24\,500$.

5 Consider the following prime factorisations:

$$32\,000 = 2^8 \times 5^3$$

$$800 = 2^5 \times 5^2$$

Find the highest common factor of $32\,000$ and 800.

6 Consider the numbers 234 and 108.

a Write the prime factorisation of 234 in expanded form.

b Write the prime factorisation of 108 in expanded form.

c Find the highest common factor of 234 and 108.

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7 Find the highest common factor of the following groups of numbers:

a 42 and 24

b 72 and 108

c 121 and 143

d 150 and 560

e 48, 80, and 176

f 168, 312, and 120

g 168, 224, and 280

h 150, 450, and 375

8 The table below shows the prime factorisations of several numbers in index form.

a Find the highest common factor of 2940 and 14 700.

b Find the highest common factor of 17 640 and 308 700.

c Find the highest common factor of 2940, 26 250, and 308 700.

Number	Prime factorisation
2940	$2^2 \times 3 \times 5 \times 7^2$
14 700	$2^2 \times 3 \times 5^2 \times 7^2$
17 640	$2^3 \times 3^2 \times 5 \times 7^2$
26 250	$2 \times 3 \times 5^4 \times 7$
308 700	$2^2 \times 3^2 \times 5^2 \times 7^3$

Lowest common multiple

9 Consider the following prime factorisations:

$$54 = 2 \times 3 \times 3 \times 3$$

$$36 = 2 \times 2 \times 3 \times 3$$

Find the lowest common multiple of 54 and 36.

10 Consider the following prime factorisations:

$$2200 = 2 \times 2 \times 2 \times 5 \times 5 \times 11$$

$$2750 = 2 \times 5 \times 5 \times 5 \times 11$$

Find the lowest common multiple of 2200 and 2750.

11 Find the lowest common multiple of the following groups of numbers:

a 24 and 84

b 78 and 26

c 144 and 156

d 450 and 650

e 40, 260, and 140

f 128, 32, and 8

g 75, 125, and 225

h 16, 36, and 124

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12 Consider the numbers 1575 and 1650.

a Find the highest common factor of 1575 and 1650.

b Find the lowest common multiple of 1575 and 1650.

Applications

13 Cicada species spend many years underground, emerging as a single brood every generation. Magicicada cassinii emerges every 17 years. They are hunted by the Latrodectus mactans spider, which has a 3 year breeding cycle.

a How often do these species start their breeding cycle at the same time?

b A genetic mutation changes the life cycle of the cicadas so the mutants emerge every 18 years. How often will the mutant cicada species start their life cycle at the same time as the spiders?

14 Two airlines, JetFuel and LionAir, each land planes at an airport terminal. During the day a JetFuel plane lands every 18 minutes, and a LionAir plane lands every 16 minutes.
A shuttlebus takes passengers from the terminal to the city center, departing every 21 minutes.

a How often does a JetFuel plane land and a shuttlebus depart at the same time?

b Which of these two options happens more often in a single day?

A JetFuel plane lands and the shuttlebus departs at the same time.

B LionAir plane lands and the shuttlebus departs at the same time.

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